

论文写作 Agent 测试文档

Abstract

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This study explores the application of large language models in automated academic paper writing by developing an intelligent agent (Agent) specifically designed for paper composition. The proposed Agent provides end-to-end assistance throughout the entire writing process, encompassing literature retrieval, outline generation, paragraph drafting, and formatting standardization. Experimental results demonstrate that the Agent significantly enhances writing efficiency while maintaining academic rigor in terms of both规范性 (normativity) and logical coherence. Additionally, this paper provides a Markdown template tailored for evaluating academic writing Agents, along with ready-to-use image prompts that can be directly replicated for practical implementation.

Keywords: 大语言模型, 智能体, 论文写作, 学术辅助, 自然语言处理

2. 引言

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With the rapid advancement of large language model technology, artificial intelligence has demonstrated significant potential in the field of content generation. Academic paper writing is characterized by its rigorous structure, standardized language, and stringent logical coherence, which traditionally require substantial time and effort to achieve. Given these challenges, the development of a dedicated academic paper writing agent holds substantial practical value.

The primary contributions of this study are as follows: First, we propose an architecture for an academic paper writing agent specifically tailored to scholarly contexts. Second, we implement modular functionalities including outline generation, literature citation management, and mathematical formula formatting. Third, the effectiveness and

usability of the proposed system are validated through multiple sets of comparative experiments.

3. 相关工作

3.1 基于大模型的文本生成

3.1 Text Generation Based on Large-Scale Language Models

In recent years, significant breakthroughs have been achieved in long-text generation through the development of models such as GPT and LLaMA. Consequently, numerous studies have explored the application of these advanced models to the field of scientific and technical document writing, aiming to leverage their capabilities for enhancing the efficiency and quality of such writing tasks.

3.2 智能体（Agent）在专业领域的应用

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Intelligent agents, through their capabilities in tool invocation, planning, and reflection, demonstrate a capacity to accomplish complex tasks across diverse domains. Extensive research has been conducted to explore their applications in specialized fields, including education, healthcare, and code development, where their functionalities have been systematically evaluated and validated. These studies underscore the potential of intelligent agents to enhance operational efficiency and decision-making processes within professional contexts.

4. 系统设计

4.1 整体架构

4.1 Overall Architecture

The academic paper writing agent primarily comprises the following functional modules: the user intention parsing module, which is responsible for interpreting and

analyzing user requirements; the literature retrieval and summarization module, which conducts targeted searches for relevant literature and generates concise summaries; the paper outline planning module, which formulates a structured and logical framework for the paper; the paragraph automatic generation module, which produces coherent and contextually appropriate paragraphs based on the established outline; and the format verification and beautification module, which ensures adherence to academic formatting standards and enhances the visual presentation of the document.

4.2 工作流程

4.2 Workflow

The operational procedure of the proposed system adheres to a structured four-stage process: First, users initiate the workflow by submitting specific parameters including the paper topic, formatting requirements, and target word count through the system interface. Second, the intelligent agent automatically generates a hierarchical three-level outline based on the input parameters, employing natural language processing algorithms to ensure logical coherence and academic relevance across all sections. Third, the system proceeds to generate full-text content chapter by chapter, dynamically incorporating academic references from established databases and inserting mathematical formulas where required, with all citations formatted according to the specified academic style guide. Finally, the completed manuscript is exported in either standardized Markdown format or Microsoft Word document (.docx), maintaining consistent formatting throughout while preserving all structural elements including section headings, figure captions (as demonstrated in Figure 3), and table references (see Table 2 for example).

5. 实验与结果

5.1 实验设置

5.1 Experimental Setup

The foundational models employed in this study are LLaMA-3 and GPT-3.5, which serve as the primary language model architectures for comparative analysis. The experimental task involves generating a computer science academic paper with an approximate word count of 3,000 characters, adhering to standard scholarly conventions in content organization and presentation. Four key evaluation metrics have been established to systematically assess model performance: (1) writing duration, measured as the total time required to complete the paper generation task; (2) structural integrity, evaluated through the coherence and logical organization of sections and subsections; (3) linguistic fluency, assessed via grammatical accuracy and naturalness of expression; and (4) formatting compliance, determined by adherence to academic writing conventions including citation style, section headings, and typographical norms.

5.2 结果分析

5.2 Results Analysis

The experimental results demonstrate that the paper-writing agent significantly outperforms human writers in terms of efficiency while simultaneously maintaining a high level of academic quality.

Table 1.

指标	人工写作	论文 Agent
平均耗时	3 天	45 分钟
结构完整度	82%	89%
语言通顺度	85%	92%
格式规范率	78%	95%

6. 结论与展望

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This study has successfully designed and implemented an intelligent agent tailored for academic paper writing, which is capable of providing comprehensive assistance

throughout the entire process, including the generation of paper outlines, the composition of main text, and the formatting of documents. Looking ahead, future research could focus on the integration of multi-agent collaboration frameworks to enhance the system's capacity for handling complex writing tasks. Additionally, the incorporation of real-time literature update mechanisms and feedback loops from domain experts could further improve the professionalism and reliability of the intelligent agent, thereby ensuring its sustained relevance and utility in academic settings.

可直接生成的图片（用于测试插图）

Images That Can Be Directly Generated (For Testing Illustrations)

To generate the relevant images, one can simply input the following prompt words into an image - generation tool.

图1：论文写作 Agent 系统架构图

****Figure 1: Architectural Diagram of the Academic Paper Writing Agent System****

The diagram illustrates the system architecture of an intelligent agent designed for academic paper writing, adopting a minimalist technological aesthetic with a dominant blue color scheme. Structured across four hierarchical layers—User Layer, Planning Layer, Execution Layer, and Tool Layer—the architecture employs a flowchart-style representation to emphasize modular clarity and functional interdependencies. Each layer is delineated through simplified geometric shapes connected by clean lines, ensuring readability as a scholarly illustration. The white background enhances visual contrast, while the absence of extraneous decorative elements aligns with academic conventions for diagrammatic precision. This modular design facilitates systematic comprehension of the agent's operational workflow, from user input reception to tool-assisted task execution. (Refer to Figure 1 for visual representation.)

图2：实验对比柱状图

Figure 2 presents a comparative bar chart illustrating the performance differences

between human-authored academic writing and AI Agent-generated text across four key evaluation metrics: time consumption, structural integrity, linguistic fluency, and formatting compliance. The visualization employs an academic color scheme with subdued, professional tones to ensure clarity and visual coherence, set against a clean white background for optimal readability. Each metric is represented by paired bars enabling direct comparison between human and AI-generated outputs, with all data points precisely aligned to maintain analytical rigor. The chart's minimalist design adheres to standard academic conventions, eliminating extraneous elements while emphasizing the comparative analysis through strategic use of negative space and consistent typography.

图3：智能体工作流程图

****Figure 3: Workflow Diagram of the Intelligent Agent****

This diagram illustrates the academic workflow of the intelligent agent, structured as a unidirectional process encompassing five sequential stages: user input acquisition, outline generation, paragraph composition, formatting optimization, and final output delivery. The visual representation employs minimalist technological design elements with clean, blue-toned lines to ensure clarity and alignment with academic illustration standards. Each stage is connected by directional arrows to emphasize the linear progression of tasks, while the overall aesthetic maintains a professional and scholarly tone suitable for inclusion in research publications.